Human Review: $\Pi(7, 15) = 7/3$ via Equiangular Lines

Original automatic proof: `solution_packet.pdf`
Checked date: 15 June 2026
Status: Correct, but probably minor as a standalone observation.
Verdict: The proof appears correct. It gives a valid answer to Basso's printed question by combining the Koenig–Lewis–Lin equality criterion with a standard equiangular-line configuration in $\mathbb{R}^7$. However, the mathematical novelty is likely limited if the underlying 28-line construction is regarded as standard.

Human Comments

The argument correctly identifies that the value 7/3 is obtained from the Koenig–Lewis–Lin bound at

$$(n, d) = (7, 15).$$

Indeed,

$$\frac{7}{15} + \sqrt{14 \cdot \frac{7}{15} \cdot \frac{8}{15}} = \frac{7}{15} + \frac{28}{15} = \frac{7}{3}.$$

The quoted Koenig–Lewis–Lin theorem gives the upper bound, and its equality case is characterized by the existence of d equiangular lines in $\mathbb{R}^n$. Thus it remains only to produce 15 equiangular lines in $\mathbb{R}^7$.

The packet does this by using the standard 28-line configuration in the hyperplane

$$H = \left\{ x \in \mathbb{R}^8 : \sum_{i=1}^{8} x_i = 0 \right\},$$

where the lines are represented by

$$v_{\{i,j\}} = e_i + e_j - \frac{1}{4}(1, \ldots, 1).$$

For two-element sets $A, B \subset \{1, \ldots, 8\}$, one has

$$\langle v_A, v_B \rangle = |A \cap B| - \frac{1}{2}.$$

Hence all these vectors have the same norm, and distinct lines have constant absolute inner product. Taking any 15 of these lines which span H gives the required equality case for $(n, d) = (7, 15)$.

The selected 15 lines in the packet do span H, since the seven vectors $v_{\{1,j\}}$, $2 \leq j \leq 8$, already form a basis. Therefore the proof of

$$\Pi(7, 15) = \frac{7}{3}$$

looks correct.

Mathematical Value

Although correct, this should probably be classified as a relatively minor observation rather than a substantial new result. The key geometric input is the classical 28-line equiangular configuration in $\mathbb{R}^7$, and the application to Basso's question is then immediate from the Koenig–Lewis–Lin equality criterion. In other words, the packet appears to notice a short consequence of known ingredients rather than introduce a new construction.

Recommended Follow-Up

This example suggests a good next step for the automated pipeline. Rather than stopping at the isolated value

$$\frac{7}{3},$$

it would be useful to rerun or adapt the search to look for a more systematic pattern. In particular, one should try to find other pairs (n, d) for which the Koenig–Lewis–Lin bound gives natural subsequent values, and then check whether known equiangular-line systems realize equality.

Ideally, the model should attempt to produce either:

a general form for such examples,

or

an infinite family of pairs (n, d) giving further explicit values.

Such a generalization would be much more mathematically valuable than the single observation $\Pi(7, 15) = 7/3$.

Review Outcome

Accepted as correct, but with low novelty as a standalone contribution. Recommended classification: correct short observation/consequence of standard equiangular-line theory. Recommended next action: rerun the problem with instructions to search for higher values, additional pairs (n, d), and possible infinite families.